

PRAKTIKUM SISTEM OPERASI

MODUL VI

(SEKUENSIAL & KONKUREN)



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

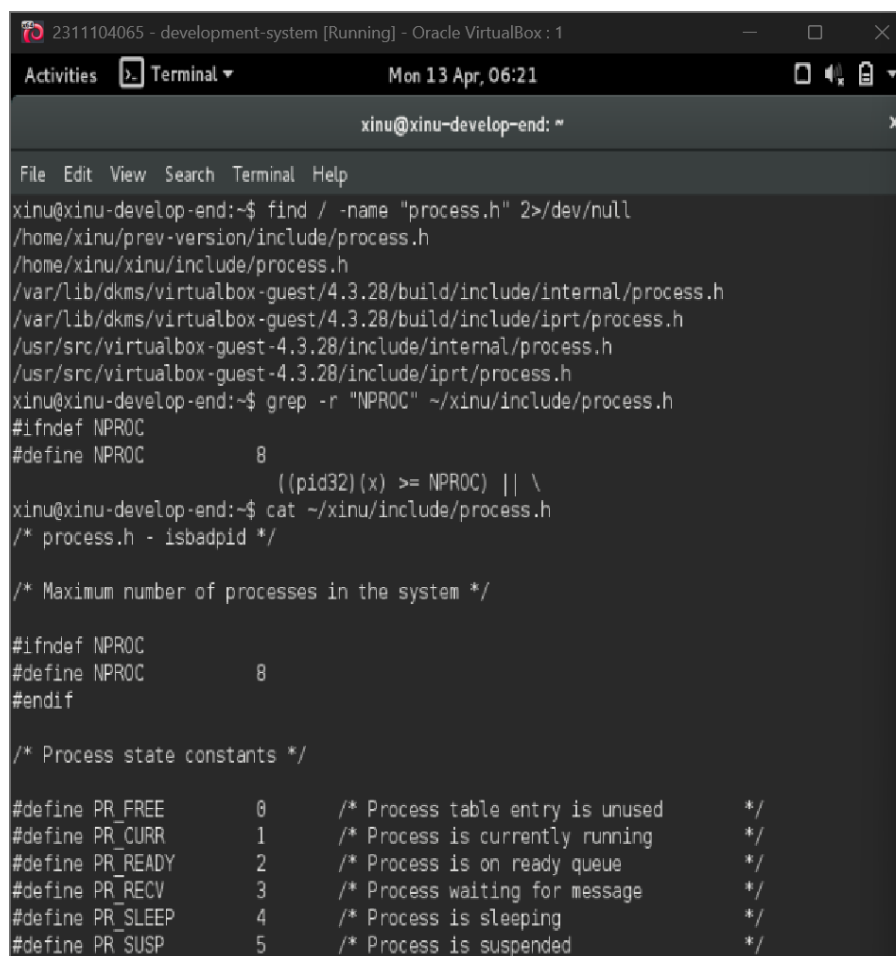
2026

JURNAL

1. Selain hardware (memory), batasan maksimal proses dapat ditentukan dengan secara software. Pada Linux maksimal proses adalah 4194303 proses (64 bit) dan 32767 proses (32 bit) dapat dilihat melalui perintah `$cat /proc/sys/kernel/pid_max` Carilah pada source code Xinu yang memberi batasan mengenai banyaknya proses yang bisa dibuat! Berapa maksimal proses dalam Xinu? Ubah menjadi maksimal 150 proses!

Jawab:

- Cari file process.h dan lihat isi file



```
2311104065 - development-system [Running] - Oracle VirtualBox: 1
Activities Terminal Mon 13 Apr, 06:21
xinu@xinu-develop-end: ~

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$ grep -r "NPROC" ~/xinu/include/process.h
#ifndef NPROC
#define NPROC      8
                ((pid32)(x) >= NPROC) || \
xinu@xinu-develop-end:~$ cat ~/xinu/include/process.h
/* process.h - isbadpid */

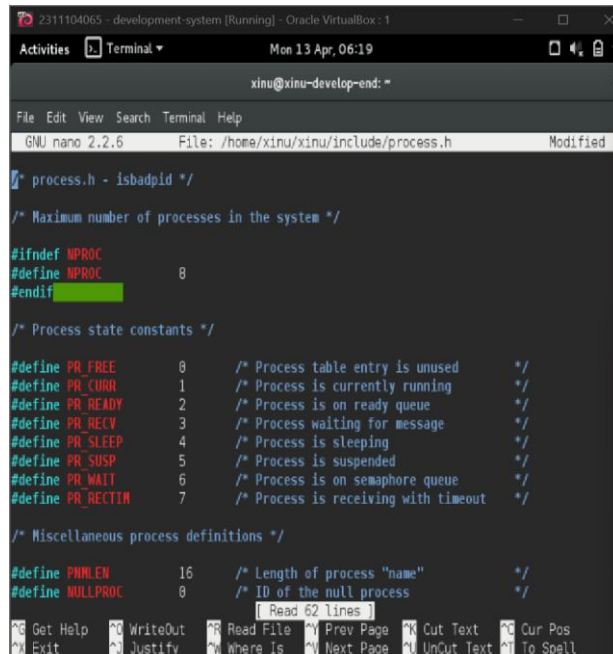
/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      8
#endif

/* Process state constants */

#define PR_FREE    0      /* Process table entry is unused      */
#define PR_CURR    1      /* Process is currently running        */
#define PR_READY   2      /* Process is on ready queue           */
#define PR_RECV    3      /* Process waiting for message         */
#define PR_SLEEP   4      /* Process is sleeping                 */
#define PR_SUSP    5      /* Process is suspended                */
```

- Ubah NPROC dari 8 menjadi 150



```

GNU nano 2.2.6 File: /home/xinu/xinu/include/process.h Modified

/* process.h - isbadpid */

/* Maximum number of processes in the system */

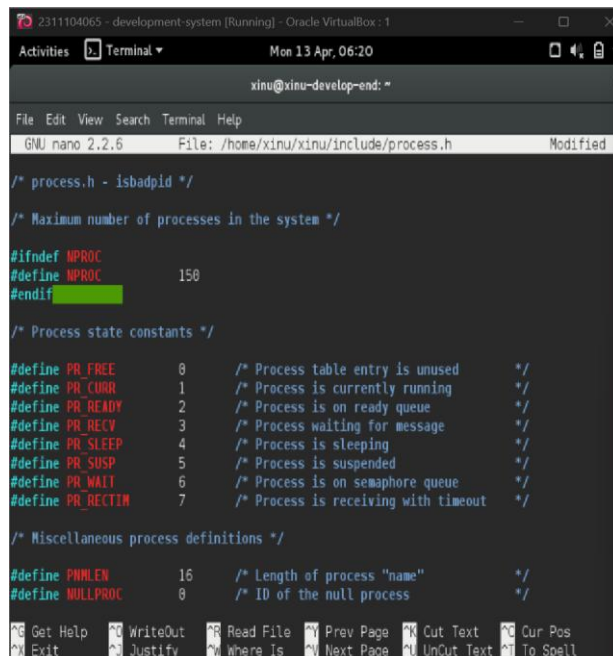
#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PWLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */
  
```



```

GNU nano 2.2.6 File: /home/xinu/xinu/include/process.h Modified

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 150
#endif

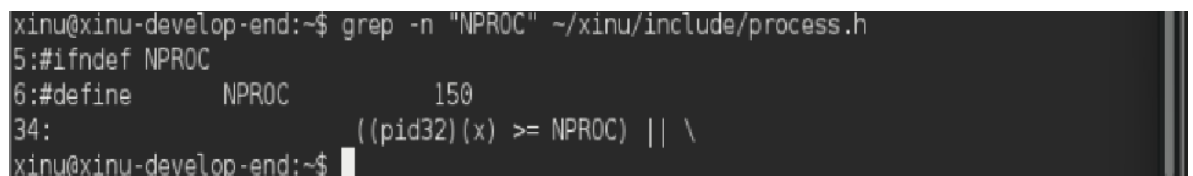
/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PWLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */
  
```

- Verifikasi perubahan



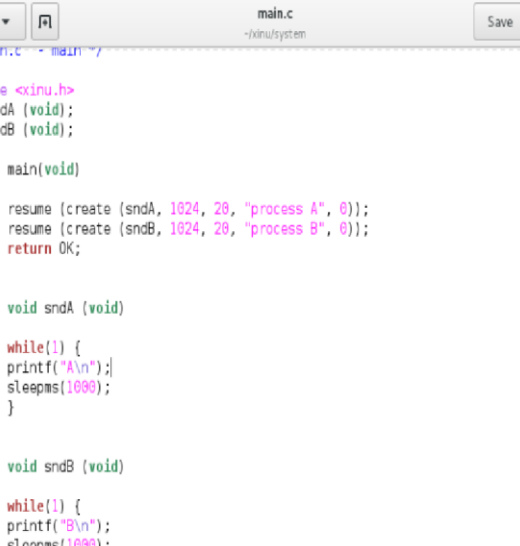
```

xinu@xinu-devel-end:~$ grep -n "NPROC" ~/xinu/include/process.h
5:#ifndef NPROC
6:#define      NPROC      150
34:      ((pid32)(x) >= NPROC) || \
xinu@xinu-devel-end:~$
  
```


3. Jalankan kode main_ex2!

Jawab:

- Syntax main_ex2



```
2311104065 - development-system [Running] - Oracle VirtualBox: 1
Activities gedit Mon 13 Apr, 06:32
Open main.c ~/vnu/system main.c Save
/* main.c ~/main */

#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    resume(create(sndA, 1024, 20, "process A", 0));
    resume(create(sndB, 1024, 20, "process B", 0));
    return OK;
}

void sndA(void)
{
    while(1) {
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1) {
        printf("B\n");
        sleepms(1000);
    }
}
```

- Outputnya

```

2311104065 - development-system [Running] - Oracle VirtualBox : 1
Activities Terminal ▾ Mon 13 Apr, 06:48
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Xinu for Vbox -- version #118 (xinu) Mon 13 Apr 09:46:31 EDT 2026

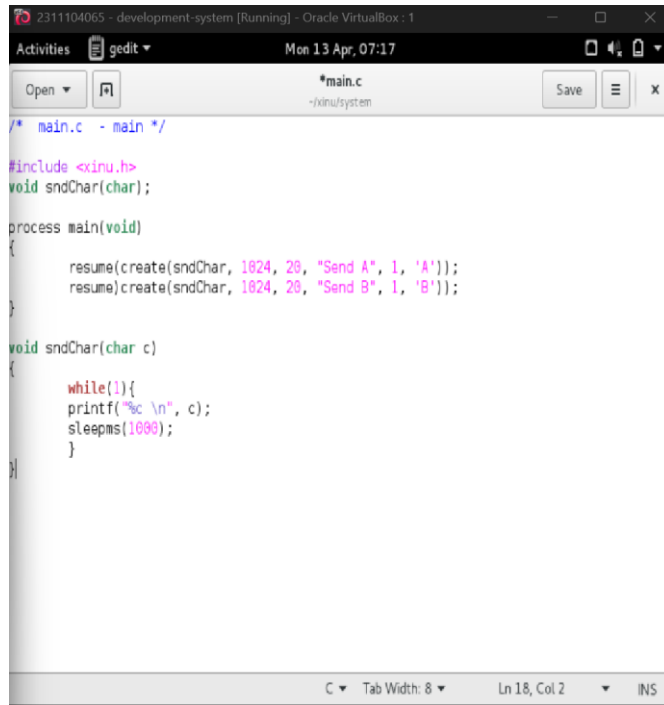
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFFF]
[0x00100000 to 0x005FBFFF]
115401 bytes of Xinu code.
[0x00100000 to 0x0011C2C8]
148456 bytes of data.
[0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
AB
B
A

CTRL-A Z for help | 115200 BNI | NOR | Minicom 2.7 | VT102 | Online 0:0 | tt
```

4. Jalankan kode main_ex3!

- Syntax main_ex3



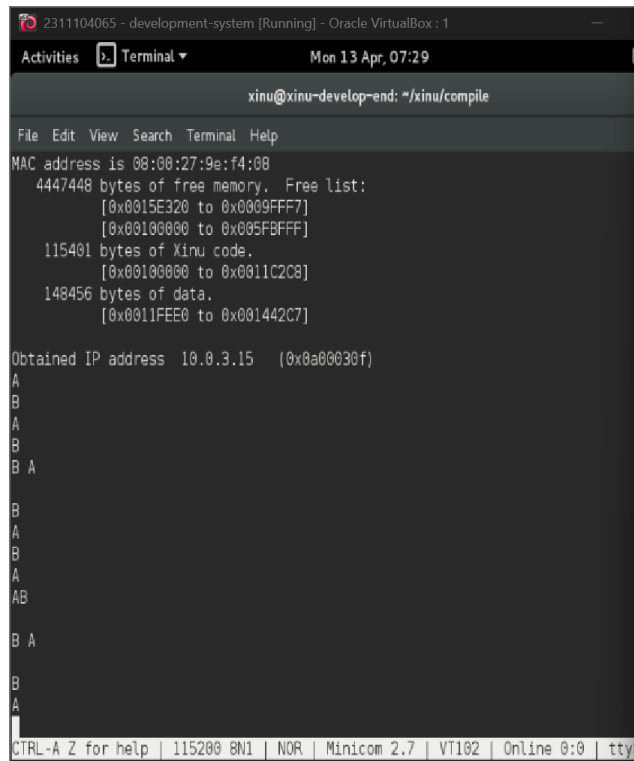
```
/* main.c - main */

#include <xinu.h>
void sndChar(char);

process main(void)
{
    resume(create(sndChar, 1024, 20, "Send A", 1, 'A'));
    resume(create(sndChar, 1024, 20, "Send B", 1, 'B'));
}

void sndChar(char c)
{
    while(1){
        printf("%c\n", c);
        sleepms(1000);
    }
}
```

- Outputnya



```
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
115401 bytes of Xinu code.
[0x00100000 to 0x0011C2C8]
148456 bytes of data.
[0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
B A
B
A
B
A
AB
B A
B
A
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS
```

5. Buatlah 2 proses produser dan konsumen. Produser memproduksi angka integer dari 1-1000. Konsumer mengkonsumsi integer yang diproduksi oleh produser dan menampilkannya! (Gunakan variabel global bertipe int32 bernama n yang digunakan secara bersama oleh kedua proses)

```
int32 n = 0; //global variabel

void produser(void){
    int32 i;
    for (i=1; i<=1000; i++){
        n++;
    }
}

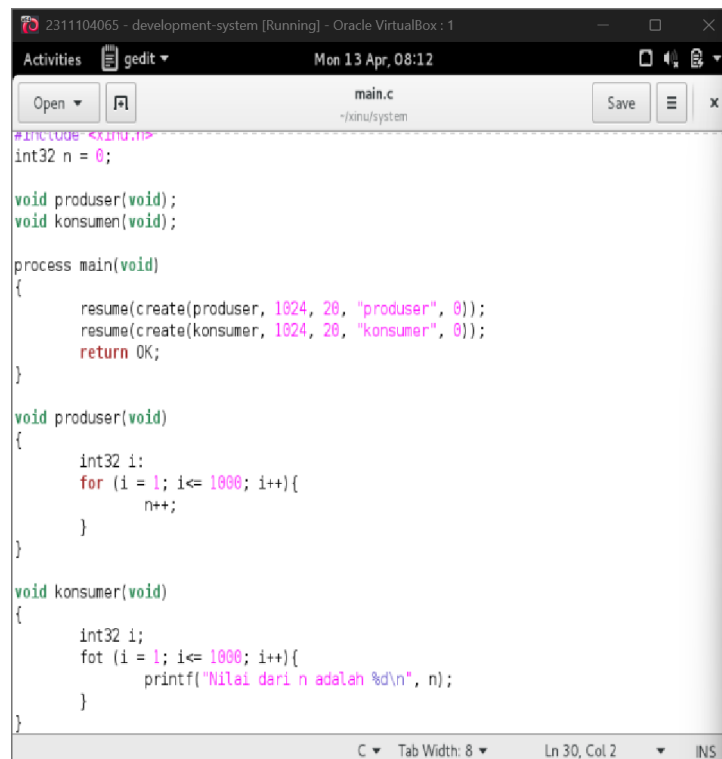
void konsumer(void){
    int32 i;
    for (i=1; i<=1000; i++){
        printf("Nilai dari n adalah %d\n",n);
    }
}
```

Hasil dari program ini cukup mengejutkan (tidak akan sesuai dengan intuisi awal).

Jelaskan mengapa hasilnya seperti itu!

Jawab:

- Syntaxnya



The screenshot shows a code editor window titled 'main.c' with the following code:

```
#include <stdio.h>
int32 n = 0;

void produser(void);
void konsumer(void);

process main(void)
{
    resume(create(produser, 1024, 20, "produser", 0));
    resume(create(konsumer, 1024, 20, "konsumer", 0));
    return OK;
}

void produser(void)
{
    int32 i;
    for (i = 1; i<= 1000; i++){
        n++;
    }
}

void konsumer(void)
{
    int32 i;
    fot (i = 1; i<= 1000; i++){
        printf("Nilai dari n adalah %d\n", n);
    }
}
```

The code is syntactically correct but contains a race condition. The producer increments a global variable 'n' from 1 to 1000. The consumer prints the value of 'n' 1000 times. Due to the lack of synchronization, the consumer may print the value of 'n' before the producer has finished incrementing it, leading to incorrect output.

- Outputnya

[illegible]

- Hasilnya tidak berurutan karena kedua proses berjalan secara konkuren dan mengakses variabel `n` secara bersamaan tanpa sinkronisasi. CPU bisa kapan saja berpindah antara proses produser dan konsumen, sehingga konsumen bisa membaca nilai `n` yang belum diupdate atau sudah dilewati oleh produser. Kondisi ini disebut *race condition*.